

Scrollytelling Portfolio · Canvas Animation & UI Design

Awwwards-inspired personal portfolio featuring HTML5 Canvas frame scrubbing, parallax text overlays, and dark/light theme system in React.

Role: Designer & Developer | Timeline: 2024 · 2025 | Tools: React 19, Vite, HTML5 Canvas, Framer Motion

01 The Creative & Technical Challenge

Most developer portfolios are either visually impressive but technically shallow, or technically impressive but visually generic. The goal was to build a portfolio where the medium is the message.

Engineered a scroll-linked HTML5 canvas animation running at 60fps with preloaded WebP frame sequencing, parallax typography layers, and instant dark/light mode switching.

02 Technical Implementation & Architecture

Canvas Rendering Pipeline: Custom requestAnimationFrame loop calculating scroll progress with smooth cubic interpolation.

Preloading & Asset Optimization: 75 WebP frames (~600KB total) preloaded asynchronously to guarantee zero jank upon user scrolling.

Responsive Stacking: Framer motion triggers calculating viewport intersection for high-density mobile and 4K desktop layouts.

03 Performance Metrics & Usability

Achieved constant 60 FPS animation frame rate during rapid inertial scrolling.

Initial page load under 2.8 seconds on simulated 3G networks.

Full WCAG AA compliance with dynamic contrast adjustment across theme toggles.